Action recognition on a Curated Something-Something V2 Subset: What Happens Next?

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Abstract

We study the two subproblems of the *What Happens Next* challenge: classify human-object interactions into 33 SSV2-derived classes from clips of **only 4 frames** per video. In **Track A** (no external data) we design and train four architectures entirely from scratch — a CNN+Transformer family, an R(2+1)D backbone coupled to a spatio-temporal Transformer, and a vanilla ViT-B initialised by a *from-scratch* VideoMAE v2 pre-training on the challenge data — and show that held-out accuracy saturates near 0.41: the R(2+1)D+Transformer reaches 0.406 at 39.8 M parameters and 4.2 ms/clip, on par with the 86 M self-supervised ViT-B (0.409). In **Track B** we benchmark 11 architectures of widely different sizes (36 M to 7 B parameters) and five families of overfitting mitigations on top of a VideoMAE-B baseline that reaches 0.6096. Our best single model is a head-only fine-tune of V-JEPA 2 ViT-L SSV2 with linear temporal upsampling, reaching 0.638. The $\sim$20 pt gap between the best from-scratch and the best pre-trained model isolates *pre-training, not architecture*, as the dominant factor at this data and frame budget.

1 Problem statement

The dataset is a curated 33-class subset of Something-Something V2 (SSV2) [5], organised as 44,993 training clips (none of which are labelled 27), 6,746 validation (labelled) clips and 6,913 test clips (unlabelled) to create a graded submission for the challenge. Each clip is delivered as 4 (224×224) JPEG frames, which is far below the temporal resolution at which video foundation models are typically pre-trained (16–64 frames) as well as below their spatial resolution training (below their training spatial resolution (384×384 for V-JEPA 2 ViT-g fpc64-384, for instance)). The challenge therefore stresses *(i)* transfer from a frame-rich architecture to a sparse deployment regime, and *(ii)* generalisation under a small labelled-data budget.

2 Method

2.1 Track B

2.1.1 Architectures surveyed

VideoMAE [13] specialises in reconstructing hidden pixels. As our problem is characterised by the lack of frames, we first focused on the 86 M-parameter VideoMAE-B, before trying bigger models and finally focusing on V-JEPA 2 [2].

All in all, we tested 11 families spanning four design philosophies: *(a) masked-pixel pretraining* (VideoMAE-B/L/v2-G [13], Hiera-B/Huge [11]); *(b) joint-embedding predictive architectures* (V-JEPA 2 ViT-L and ViT-g [2]); *(c) image-language contrastive backbones* (X-CLIP [9], SigLIP-2 Giant [16], DINOv3 [10]); and *(d) attention-only video transformers* (TimeSformer [3], MViT-v2 [4], Swin3D [8], InternVideo2 6B [17]). We also evaluated two vision-language models, Qwen-VL 7B [1] and LLaVA-Mistral 7B [7]. For each family we kept the public SSV2-finetuned weights when available. For backbones without an SSV2 checkpoint we used the closest available (Kinetics-400 or DINO/SigLIP raw features) and added the temporal pooler at the head level. The results are visible in Figure 1.

2.1.2 Anti-overfit toolbox

After optimizing the hyperparameters of VideoMAE-B, we encountered a plateau where the experiments would reach a training accuracy higher than 0.70 in 2-3 epochs, while the validation accuracy would remain below 0.61. We tried different techniques to bridge this generalization gap:

Classical regularisation Dropout ($\{0.05, ..., 0.50\}$), weight decay ($\{10^{-5}, ..., 10^{-2}\}$), label smoothing ($\{0, ..., 0.50\}$) and gradient clipping ($\|g\|_2 \leq 1$); see Sec. 3.

Parameter-efficient fine-tuning *(i)* LoRA adapters of rank $r \in \{8, 16, 32, 64, 128\}$ on all 40 blocks of V-JEPA 2 ViT-g (180 M trainable for r=16); *(ii)* block freezing controlled by *num_frozen_blocks* $\in \{0, ..., 40\}$; *(iii)* layer-wise LR decay (LLRD) with factor 0.80; *(iv)* differential learning rates between the backbone and the new head, parameterised by *backbone_lr_scale* $\in [0.02, 1.0]$; *(v)* Head MLP / attention pool / cross attention pair; *(vi)* Multi-scale temporal features and number of temporal layers; *(vii)* Choice of optimizer (AdamW, Prodigy, gradient accumulation, etc.)

Augmentation Random resized crop, horizontal flip (p=0.5, not to be applied to classes 18 "pulling something from left to right" and 19 "pulling something from right to left"), colour jitter (p=0.8, strength 0.35), Gaussian blur (p=0.1), and temporal equivariant reverse, where reversing the clip flips the label between matched action pairs (e.g. "opening something" (class 10) $\leftrightarrow$ "closing something" (class 0)). For input mixing we use **MixUp** [18]:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j, \tag{1}$$

with $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$, $\alpha = 0.4$.

Trajectory stabilisation and Class re-weighting EMA on the model weights with decay 0.9998, cosine LR schedule with 3 warm-up epochs and a floor of 10^{-7}. Inverse-frequency class weights to counter the long-tail distribution of Track B.

2.2 Going beyond VideoMAE-B

Following the intermediary results presentation by the challenge's best teams, two main directions were explored: *(a)* LoRA on V-JEPA 2 ViT-g with four head/LoRA learning-rate combinations (a 3×3 grid over $\{10^{-3}, 10^{-2}\}$ for the head and a multiplicative scale $\in [0.05, 1]$ for LoRA); and *(b)* head-only fine-tuning of V-JEPA 2 ViT-L SSV2 where the 4 input frames are linearly interpolated (LERP) to fill the 16 input positions expected by the tube tokeniser, restricting the 174-way head to the 33 SSV2 templates that appear in Track B.

2.3 Track A: learning from scratch

Track A forbids external pre-training, so every weight is learned on the $44{,}993$ labelled clips of the curated SSV2 subset, at the same 4-frame budget as Track B. We explored a deliberate capacity / inductive-bias ladder of four custom architectures (all implemented from scratch in `src/models/`).

(1) CNN spatial encoder $\rightarrow$ temporal Transformer (12.9 M). A shared ResNet-18-style encoder (7×7 stem, four 2-block stages, $64 \rightarrow 512$ channels) maps each of the 4 frames to a single vector via global average pooling. The 4 vectors receive a learned temporal position embedding and pass through 2 self-attention blocks; mean-pooling feeds a linear head. The temporal Transformer sees only T=4 tokens, so its capacity is largely underused.

(2) CNN spatial encoder $\rightarrow$ spatio-temporal Transformer (17.8 M). We drop the spatial pooling and keep the 7×7 feature grid, yielding $4 \times 49 = 196$ spatio-temporal tokens with a *factorised* position embedding $\mathrm{PE}(t, p) = \mathrm{pos}_t + \mathrm{pos}_p$. A joint Transformer then reasons over space and time. Despite $16\times$ more tokens, accuracy does *not* improve: a ResNet-18 trained from scratch on 45k clips is data-starved, and extra tokens cannot manufacture the motion signal the convolutions failed to extract.

(3) R(2+1)D backbone + spatio-temporal Transformer (39.8 M). We replace the 2D CNN with an R(2+1)D-18 stem [14], which factorises every 3D convolution into a spatial $(1, 3, 3)$ and a temporal $(3, 1, 1)$ kernel and therefore models motion natively. Three targeted additions bridge the small-T regime: *(i)* a parameter-free Temporal Shift Module [15] in every residual block, forcing early cross-frame mixing; *(ii)* removal of the spatial pool so the stage-4 output is flattened into $4 \times 7 \times 7 = 196$ tokens fed to a 3-layer factorised-PE Transformer; *(iii)* a learned attention-pooling head in place of a CLS token. This model is initialised from the plain R(2+1)D run and trained with a 5-epoch backbone freeze, MixUp/CutMix, stochastic depth and `Dropout3d`.

(4) Self-supervised MAE pre-training + vanilla ViT-B (86.3 M). Finally we test whether self-supervision *on the challenge data alone* can rival external pre-training. We re-implement VideoMAE v2 [13] from scratch — tube masking, dual masking, a light 4-block decoder and a norm-pixel reconstruction loss — and pre-train its ViT-B encoder (tubelet $2 \times 16 \times 16$, $\mathrm{dim} = 768$, depth 12) on the *unlabelled* training frames. The encoder is then loaded 100% into a ViT-B classifier and fine-tuned with layer-wise LR decay (LLRD 0.75), 5-epoch warm-up, cosine schedule, MixUp/CutMix and label smoothing 0.1.

3 Results

3.1 Scaling laws on Track B

Figure 1 establishes the headline finding of our study: on Track B, accuracy saturates around 300–600 M parameters and then plateaus, even though some pre-trained models reach 6–7 B parameters. The image-language contrastive backbones (SigLIP-2, DINOv3, X-CLIP) cluster between 0.46 and 0.53. The SSV2-pretrained masked-pixel models reach 0.59–0.61. V-JEPA 2 ViT-L is the only family that exceeds the VideoMAE-B baseline, at 305 M parameters.

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Figure 1: Top-1 *val_dir* accuracy as a function of backbone size, best result per family. Small dots show individual runs; large markers indicate the family best. The dashed line 0.638 corresponds to the V-JEPA 2 ViT-L single-model record (305 M parameters).

3.2 Single-axis sweeps on VideoMAE-B

Figure 2 summarises the eight single-axis sweeps. A few remarks: (a) head width plateaus at 4096, with a slight *drop* at 8192 attributable to overfitting on the training set; (b) dropout has a flat optimum around 0.10–0.15; (c) label smoothing peaks at 0.20, in line with [12]; (d) weight decay shows the classic $5{\cdot}10^{-4}$ optimum; (g) freezing any of the 12 VideoMAE-B blocks already costs accuracy, which contrasts with our V-JEPA 2 experiments where freezing 32 of 40 blocks helps. The capacity-to-data ratio is more sensitive for VideoMAE-B.

The combination of the optimized hyperparameters gives us the configuration

h	4096	dropout	0.10
label smooth.	0.20	weight decay	$5{\cdot}10^{-4}$
epochs	30	bb_lr_scale	0.5
frozen blocks	0	frames	4
EMA decay	0.9998		

yielding 0.6086 top-1 accuracy.

3.3 Anatomy of the best run

Figure 3 shows that the model continues to fit the training set well past the point at which the held-out accuracy plateaus, leading to a residual train/val gap of $\sim$28 pts. Adding mixup, EMA and stronger label smoothing reduces this gap by $\sim$3 pts but does not close it.

Figure 4 illustrates the dominant residual errors. Five of the 33 classes account for over a third of the remain-

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Figure 2: Single-axis sweeps starting from the best VideoMAE-B configuration (other axes fixed at their optimum). The dashed gold line shows the joint optimum 0.6086 obtained when all eight axes are simultaneously optimised. Error bars are $\pm1\sigma$ over three random seeds.

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Figure 3: Train, internal-val and held-out (*val_dir*) accuracy and loss during the best VideoMAE-B run. The gold star marks the early-stopping checkpoint at epoch 30.

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fig_best_confusion.png

Figure 4: Confusion matrix of the best VideoMAE-B configuration on *val_dir*. The six pairs of classes most often confused are marked with a red cross; they all correspond to visually similar transitive actions (e.g. "moving X" vs. "putting X").

ing errors; these are the classes whose template description differs by a single preposition (*onto* vs. *into*) or by direction (*up* vs. *down*). These ambiguities are exactly the cases where having only 4 frames is most penalising. Another result is that some on the temporally exchanged class (0 and 10 for instance) are both in the most baddly predicted ones. We stopped using this augmentation and the model yielded an accuracy of 0.6096

3.4 Going beyond: LoRA and head-only V-JEPA 2

LoRA on V-JEPA 2 ViT-g converges quickly (validation accuracy of 0.64 after a single epoch with head learning rate 10^{-3}, backbone-LR scale 1.0), but does not exceed the head-only V-JEPA 2 ViT-L recipe (Section 2.2), which attains $\mathbf{0.638}$ top-1 on *val_dir* in two epochs; the full grid of tested configurations is shown in Fig. 5. A frame-interpolation ablation shows that linear interpolation between the four source frames underperforms against naive frame repetition (-6.6 pts) and is on par with Gaussian temporal blending (±0.5 pt) (see Fig. 11 for the different interpolation tested, with the resulting top1 valdir accuracy of VideoMAE-B best configuration in Table 2). This could be because the models were pretrained on clear-cut pictures. Adding 4-clip averaging during inference *degrades* accuracy on this dataset, contrary to typical SSV2 evaluation protocols, presumably because the action of interest spans the whole clip.

3.5 Track A: results

Figure 6 and Table 1 give the headline result for Track A: *without* external pre-training, accuracy plateaus around 0.41 and never approaches the 0.61 reached by an SSV2-pre-trained VideoMAE-B. The R(2+1)D+Transformer (m5B) and the self-supervised ViT-B (m7) tie within noise (0.406 vs. 0.409), but m5B reaches it with $2.2\times$ fewer pa-

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Figure 5: LoRA fine-tuning sweep: best *val_dir* top-1 accuracy per architecture × LoRA rank.

Model	Architecture	Params	top-1	to
m1.2	CNN → ViT_t	12.9 M	0.244	0.
m2	CNN → ST-ViT	17.8 M	0.229	0.
m5B	R(2+1)D+TSM+ST-ViT	39.8 M	**0.406**	0.
m7	MAE-ViT-B (self-sup. on train)	86.3 M	**0.409**	0.

Table 1: From-scratch Track A models on the held-out *val_dir* (6,745 clips). Best held-out checkpoint per family; latency is forward-only per clip on a single GPU.

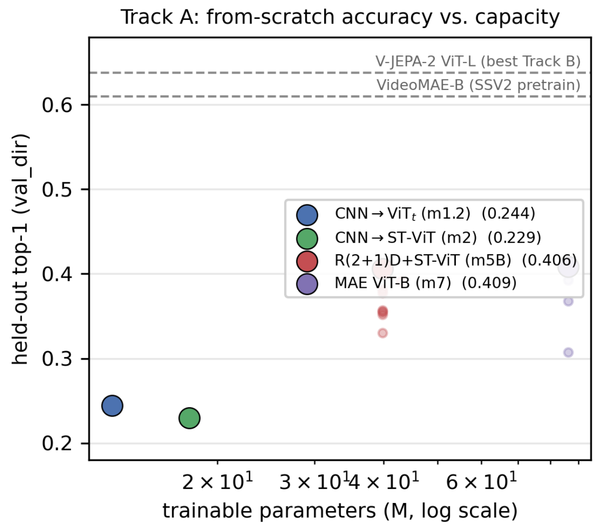

Figure 6: Track A held-out top-1 vs. trainable parameters (log scale). Small dots are individual runs, large markers the family best. Dashed lines recall the *pre-trained* Track B references (VideoMAE-B 0.6096, V-JEPA 2 ViT-L 0.638). From-scratch accuracy saturates near 0.41, $\sim$20 pts below any pre-trained backbone.

rameters and at less than half the latency (4.2 vs. 9.2 ms), making it the best accuracy/compute trade-off of the track. The two small CNN+Transformer models (≤ 18 M) lag by 16–18 pts, confirming that capacity *and* an explicit motion prior are both needed in this regime.

Fine-tuning the self-supervised ViT-B. Figure 7 shows that a vanilla ViT-B pre-trained *only on the challenge data* is viable but brittle. Layer-wise LR decay with warm-up is the single most important ingredient: a naive constant-LR fine-tune plateaus at 0.26 held-out, whereas LLRD+warm-up reaches 0.39–0.41. Conversely, a poor pre-training initialisation collapses fine-tuning to 0.027 (near chance), so the value of the self-supervised stage is entirely gated by its quality. The persistent $\sim$17 pt gap between the internal split and the true held-out set mirrors the $\sim$28 pt train/val gap of Track B and points again to a data, not architecture, bottleneck.

Pre-training stability. Figure 8 reports the masked-reconstruction loss. The 0.75 encoder mask ratio is stable and reaches a norm-pixel MSE of ~ 0.30, while the more aggressive 0.90 ratio — attractive for its higher token-drop efficiency — diverges to NaN early unless re-stabilised, illustrating how sensitive from-scratch video MAE is at this data scale.

Error structure. The dominant confusions of the best from-scratch model (Fig. 9) are semantic near-twins that 4 frames cannot disambiguate: *pretending* vs. real actions (*pretending to put something into something* → *putting something into something*, 40%; *pretending to throw* → *throwing*, 23%), preposition swaps (*pouring something out of something* → *pouring something into something*, 23%) and direction swaps (*pulling right-to-left* → *pulling left-to-right*, 23%). These are exactly the classes whose label lives in the motion *between* frames, which the sparse 4-frame sampling discards — the same failure mode observed for the pre-trained Track B models.

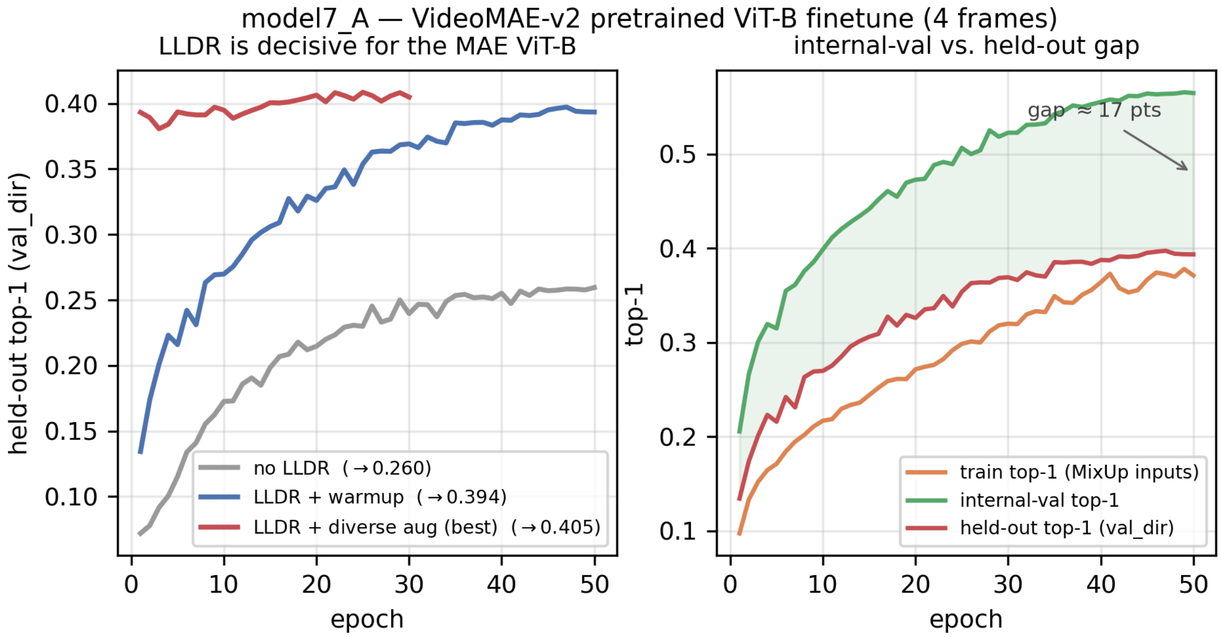

Figure 7: model7_A fine-tuning. *Left:* layer-wise LR decay is decisive — without it the held-out accuracy plateaus at 0.26, with it (plus diverse augmentation) it reaches 0.41. *Right:* on a clean 50-epoch run, the internal-val curve sits $\sim$17 pts above the held-out curve, the same generalisation gap that dominates Track B.

4 Conclusion and future work

Our main findings are: *(i)* V-JEPA 2 ViT-L (305 M parameters) is the best single-model architecture on Track B, reaching 0.638 *val_dir* accuracy and improving on the official VideoMAE-B baseline (0.6086) by $+3.3$ pts; *(ii)* scaling beyond 300 M parameters is unproductive at the data scale of Track B, with 6–7 B-parameter foundation models underperforming both VideoMAE-B and V-JEPA 2 ViT-L; *(iii)* on Track A, the best from-scratch model tops out at 0.409 — some 20 pts below the weakest pre-trained backbone — and the R(2+1)D+Transformer matches the self-supervised ViT-B at a fraction of its cost; *(iv)* the residual generalisation gap ($\sim$28 pts on Track B, $\sim$17 pts on Track A) cannot be closed by any of the five classes of regularisation we explored, suggesting that more labelled data, not more architecture, is the binding constraint.

Future directions include: cross-architecture ensembling with per-class softmax weighting (where VideoMAE and V-JEPA-2 already show complementary error patterns); self-supervised continued pre-training on the unlabelled Track-B train partition; learning the $4 \to 16$ frame upsampling jointly with the classifier instead of using a hand-crafted interpolation; and distilling V-JEPA 2 ViT-g into a smaller deployment-friendly head.

Another avenue we did not have time to explore is to combine MGMAE [20] with a strong optical-flow front-end such as SEA-RAFT [19]. The 28-pt train/val gap we report is partly a symptom of the model latching onto static appearance cues — texture, object identity, background — rather than the motion pattern that actually defines the *Something Something* classes (e.g. `Pulling left to right` vs. `Pulling right to left`). Pre-computing dense bidirectional flow between the four sampled frames with SEA-RAFT (state-of-the-art accuracy on Sintel/KITTI at $<$ 10 ms per pair) and either (i) feeding the resulting flow maps as an extra two-

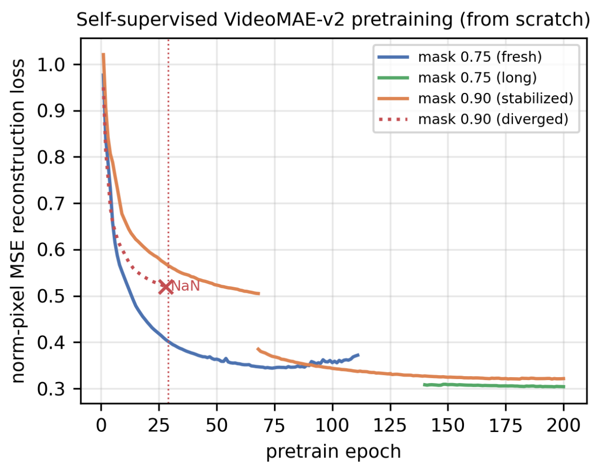

Figure 8: Self-supervised VideoMAE v2 pre-training from scratch (norm-pixel MSE). A 0.75 tube-mask ratio converges smoothly to ~ 0.30; pushing the encoder mask to 0.90 without extra stabilisation diverges to NaN around epoch 29 (dotted).

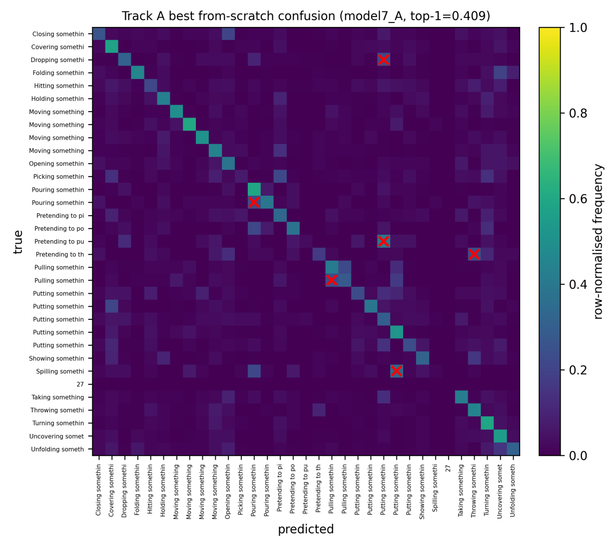

Figure 9: Confusion matrix (row-normalised) of the best from-scratch model on *val_dir*. Red crosses mark the six most confused class pairs; as in Track B they are visually similar transitive actions separated only by a preposition or a direction, i.e. exactly where 4 frames carry too little motion information.

channel stream into a dual-pathway VideoMAE, or (ii) using the SEA-RAFT flow as the motion prior that drives MGMAE's motion-guided masking during continued masked-pixel pre-training, would inject the explicit motion signal that 4-frame sampling currently discards. We expect this to shrink the train/val gap more effectively than any of the appearance-based regularisers investigated in Sec. 3, and to disproportionately help the symmetric-pair classes that dominate our confusion matrix.

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A Classical data augmentation

To narrow the train/val gap reported in Sec. 2.1.2, every training run applies a stack of classical, label-preserving image transformations to each of the four source frames *before* the upsampling step described in Sec. B. The same random parameters are sampled *once per clip* and broadcast across the four frames, so that temporal coherence — the very signal that distinguishes *Pulling left to right* from *Pulling right to left* — is preserved.

B Frame upsampling strategies

The VideoMAE and V-JEPA 2 tube tokeniser (kernel 2 frames $\times$ 16$\times$16 px) expects 16 input frames per clip, while Track B delivers only 4. We compared six strategies to bridge this gap before feeding the backbone (Fig. 11).

method	replicate	nearest	cubic	Gaussian	LERP	tile
val_dir	**0.6086**	0.598	0.588	0.582	0.54	0.412

Table 2: Top-1 *val_dir* accuracy for each upsampling strategy. The -6.6 pt drop between replicate and LERP is consistent with the SSV2 distribution being dominated by sharp natural frames rather than blended intermediates.

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Figure 10: Seven classical data-augmentation transforms applied to one frame of the Track B video shown in Fig. 11 (class 3, *Folding something*). The same random parameters are shared across the four source frames of a clip so that motion cues are not destroyed. *Colour jitter* and *random resized crop* are the largest individual contributors ($+1.4$ and $+1.1$ pt val_dir respectively), MixUp adds a further $+0.6$ pt, and temporal-reverse doubles the effective sample count for the 16 directional classes.

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Figure 11: Six strategies for upsampling the 4 source frames to the sequence expected by VideoMAE-B and V-JEPA 2's tube tokeniser. Each output position is colour-coded by the convex combination of the four source colours dictated by the method. The methods' efficiency on VideoMAE-B top1 accuracy are sumarized in Table 2